Digital Filtering DSAP

digital filtering in detail with image in dsap

2.1.2 Digital Filtering (DSAP) – Detailed Explanation

English + Nepali Mixup (No Romanize)

Illustration

Introduction

Digital Filtering भनेको **Digital Signal Processing (DSP)** को एउटा महत्वपूर्ण process हो जसले **desired signal लाई preserve** गर्दै **unwanted noise, interference, वा unwanted frequency components हटाउने** काम गर्छ।

Simple शब्दमा,

> **Digital Filter = एउटा mathematical system जसले input digital signal लाई process गरेर better output signal produce गर्छ।**

Analog filter मा resistor, capacitor, inductor प्रयोग हुन्छ भने Digital filter मा mathematical calculations प्रयोग हुन्छ।

Digital filtering computer, DSP processor, FPGA, microcontroller वा software द्वारा गरिन्छ।

Definition

Digital Filtering is the process of modifying a digital signal to remove unwanted components and enhance desired information using mathematical algorithms.

नेपाली अर्थ:

Digital filtering भनेको mathematical algorithm प्रयोग गरेर digital signal बाट आवश्यक information राख्ने र unwanted part हटाउने प्रक्रिया हो।

Basic Block Diagram

```
Input Signal
     |
     ▼
+----------------+
| Digital Filter |
+----------------+
     |
     ▼
Filtered Output
```

Real Signal Example

Suppose microphone ले voice record गरिरहेको छ।

Input Signal

```
Voice + Fan Noise + Background Noise
```

Digital Filter पछि

```
Clean Voice
```

यसले unwanted fan noise remove गर्छ।

Why Digital Filtering is Needed?

Digital filtering धेरै कारणले प्रयोग हुन्छ।

1. Noise Removal

Signal मा unwanted noise हुन्छ।

Example

```
Voice + Noise
```

↓

```
Voice
```

2. Improve Signal Quality

Signal clearer र smoother हुन्छ।

Image processing मा पनि pixels improve गर्न प्रयोग हुन्छ।

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3. Remove Interference

Power line interference

50 Hz

60 Hz

Radio interference

EMI

यस्ता unwanted frequencies हटाउन।

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4. Feature Extraction

Speech recognition

Face detection

Medical diagnosis

Radar

Communication

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5. Data Compression Preparation

Noise हटाएपछि compression अझ राम्रो हुन्छ।

Working Principle

Digital filter discrete-time samples मा काम गर्छ।

Continuous Signal

```
Analog

~~~~~~~~~~~~~~
```

Sampling पछि

```
● ● ● ● ● ● ●
```

Digital Filter

↓

Output Samples

```
○ ○ ○ ○ ○ ○ ○
```

Mathematical Representation

Input

```
x(n)
```

Output

```
y(n)
```

Digital filter

```
y(n)=Filter[x(n)]
```

FIR example

```
y(n)=0.2x(n)+0.5x(n−1)+0.3x(n−2)
```

यसले तीनवटा previous samples प्रयोग गरेर output calculate गर्छ।

Types of Digital Filters

Digital filters मुख्य दुई प्रकारका हुन्छन्।

1. FIR Filter

Full Form

Finite Impulse Response

Characteristics

- Finite output duration
- Always stable
- Linear phase possible
- Easy to design
- Slightly slower

Equation

```
y(n)=Σ bk x(n-k)
```

2. IIR Filter

Full Form

Infinite Impulse Response

Characteristics

- Feedback प्रयोग गर्छ
- Faster
- Less memory
- More efficient
- Can become unstable if not designed properly

Equation

```
y(n)=Σ bk x(n-k) − Σ ak y(n-k)
```

Classification According to Frequency

---

1. Low Pass Filter (LPF)

Pass गर्छ

Low frequencies

Blocks

High frequencies

Example

Audio smoothing

Noise reduction

Illustration

```
Input

^^^^^^^^^^^^
~~~~~~~

Output

~~~~~~~
```

2. High Pass Filter (HPF)

Pass गर्छ

High frequencies

Blocks

Low frequencies

Used for

Edge detection

Image sharpening

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3. Band Pass Filter (BPF)

Pass गर्छ

Only selected frequency range

```
-----████-----
```

Example

Radio tuning

Wireless communication

---

4. Band Stop Filter (Notch Filter)

Remove गर्छ

Specific frequency

Example

50 Hz power noise remove

```
████----████
```

Digital Filtering Process

Step 1

Receive sampled signal

↓

Step 2

Apply mathematical algorithm

↓

Step 3

Calculate new sample values

↓

Step 4

Generate filtered output

Example

Original Signal

```
Voice + Noise
```

```
**************
~~~~Voice~~~~~
**************
```

After Filter

```
~~~~Voice~~~~~
```

Noise हट्यो।

Real Life Applications

Mobile Phone

Call गर्दा surrounding noise हटाउन।

WhatsApp Voice Call

Background noise remove।

Echo cancellation।

YouTube

Audio enhancement।

Video noise reduction।

Hospital

ECG signal filtering।

EEG analysis।

MRI image enhancement।

Communication System

Receiver ले noise हटाउँछ।

Better data recovery।

Radar

Object detect गर्न।

Noise remove।

Satellite Communication

Weak signals improve।

Image Processing

Blur remove।

Sharpening।

Edge enhancement।

Speech Recognition

Google Assistant

Siri

Alexa

Noise हटाएर speech detect गर्छ।

Music Processing

Equalizer

Bass control

Treble control

Noise reduction

Advantages

- Removes unwanted noise.
- Improves signal quality.
- Highly accurate.
- Programmable and flexible.
- Stable (especially FIR filters).
- Easy to modify using software.
- Suitable for real-time processing.
- High reliability.
- Can process multiple signals efficiently.

Disadvantages

- Requires computation power.
- Processing delay may occur.
- High-speed processors are needed for complex filters.
- Design can be mathematically complex.
- IIR filters may become unstable if improperly designed.

Analog Filter vs Digital Filter

Analog Filter	Digital Filter
Continuous signal process गर्छ	Digital signal process गर्छ
Uses electronic components	Uses mathematical algorithms
Hard to modify	Easy to modify
Lower flexibility	Highly flexible
More affected by component variations	More accurate and repeatable
Hardware dependent	Software programmable

Advantages of FIR over IIR

- Always stable.
- Linear phase response possible.
- No feedback, making implementation simpler.
- Preferred where signal phase must be preserved.

Advantages of IIR over FIR

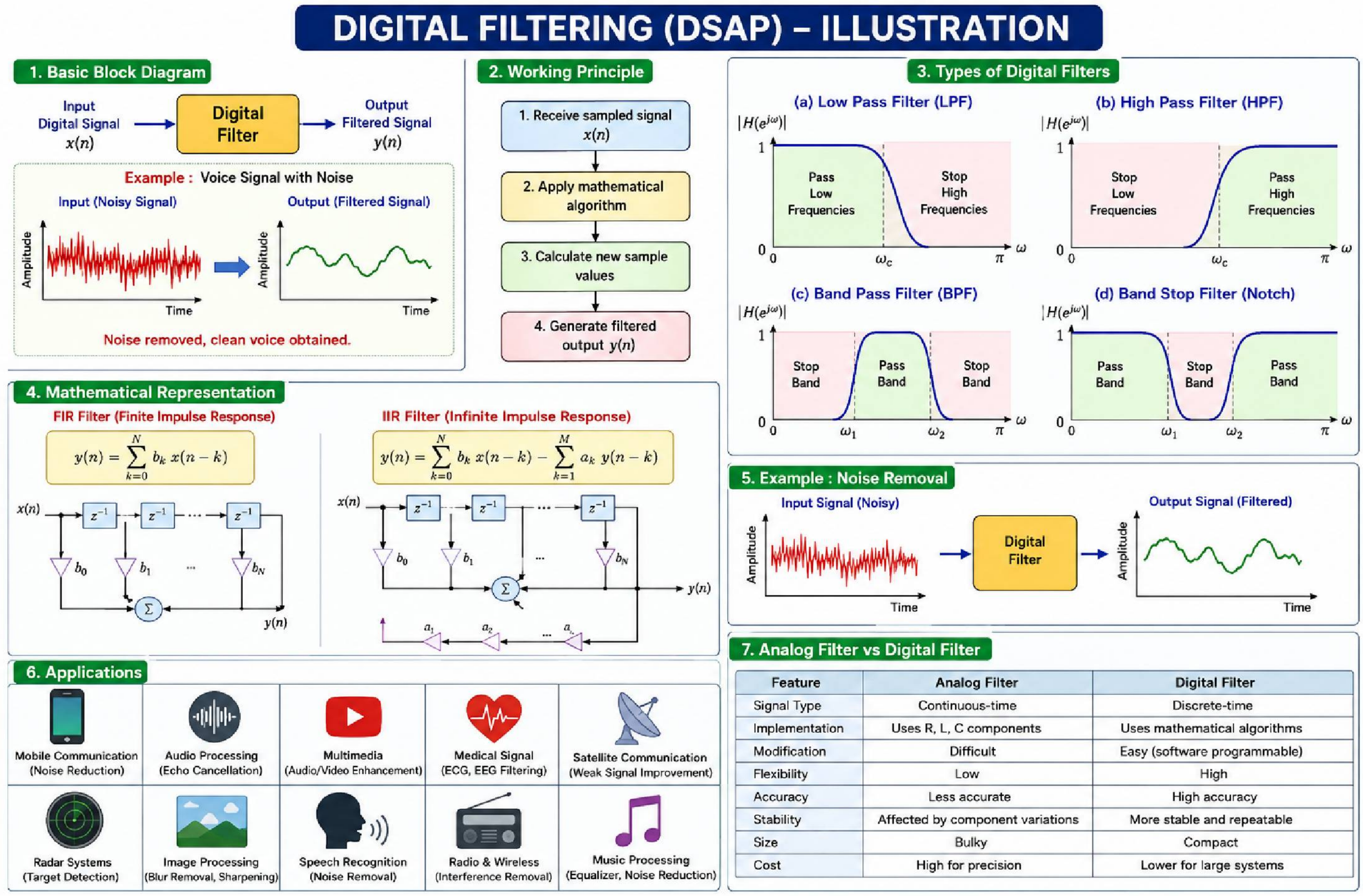
- Requires fewer coefficients.
- Faster computation.
- Lower memory requirement.
- Efficient for many real-time applications.

Summary

Digital Filtering DSP को एउटा अत्यन्त महत्वपूर्ण technique हो जसले **digital signal बाट unwanted noise तथा interference हटाएर desired information सुरक्षित राख्छ।** यो mathematical algorithms प्रयोग गरेर काम गर्छ र **FIR** तथा **IIR** दुई प्रमुख प्रकारमा विभाजन गरिन्छ। Frequency अनुसार **Low Pass, High Pass, Band Pass,** र **Band Stop (Notch)** filters प्रयोग गरिन्छ। Digital filtering को प्रयोग **mobile communication, audio processing, image processing, medical instruments, radar, satellite communication, speech recognition,** र **multimedia systems** मा व्यापक रूपमा गरिन्छ।

illustration image generate

OpenAI



Feature	Analog Filter	Digital Filter
Signal Type	Continuous-time	Discrete-time
Implementation	Uses R, L, C components	Uses mathematical algorithms
Modification	Difficult	Easy (software programmable)
Flexibility	Low	High
Accuracy	Less accurate	High accuracy
Stability	Affected by component variations	More stable and repeatable
Size	Bulky	Compact
Cost	High for precision	Lower for large systems